Two endpoint anchors on seven points

Human Compression project

Lemma 1. *Let a, b, c be involutions on a set X of seven points, with $ac = ca$. Suppose that distinct points $u, v \in X$ satisfy*

$$a(u) \neq u, \quad b(u) = c(u) = u, \qquad c(v) \neq v, \quad a(v) = b(v) = v.$$

Then $\langle a, b, c\rangle$ is intransitive on X.

Proof. Put $K = \langle a, c\rangle$. Since a and c commute and have order two, every K-orbit has size 1, 2 or 4. The orbit of u is exactly $\{u, a(u)\}$: indeed, c fixes both points because $c(a(u)) = a(c(u)) = a(u)$. Similarly, the orbit of v is exactly $\{v, c(v)\}$. These two orbits are distinct. Otherwise v would belong to the first orbit, on which a interchanges its two points, contrary to $a(v) = v$.

These orbits account for four points. The other three cannot form a single K-orbit, since three is not an allowed orbit size. Hence K has at least four orbits on X.

The involution b fixes u and v. It is therefore a product of at most two disjoint transpositions on the other five points. Start with the graph whose connected components are the K-orbits, and add an edge for each transposition of b. Each added edge reduces the number of components by at most one. There remain at least $4 - 2 = 2$ components. They are precisely the orbits of $\langle a, b, c\rangle$, since paths using the edges of the three involutions describe their generated orbits. Thus the generated group is intransitive. □

Application and scope. A rank-three string representation on seven points with anchors at its first and last marks satisfies the hypotheses. It cannot generate the natural symmetric group. The conclusion requires neither the intersection property nor two eligible edges for each mark. It resolves only this seven-point boundary; it makes no assertion about higher degrees.

Verification and attribution. The written proof is self-contained. A separate enumeration normalizes $u = 1, v = 7$ by relabelling. It examines all 231 nonidentity involutions, then all 200 commuting endpoint pairs and 5000 anchored triples. None is transitive. An independent GAP enumeration using actual generated group orbits gives the same counts. Positive controls in degrees four and eight, and a seven-point control missing one anchor condition, show that these tests do not merely reject every input. These computations supplement the proof; they are not its completeness premise. The requested boundary was specified by the investigator of the critical endpoint argument. The orbit proof and the two enumerations were developed here. No priority, whole-manuscript, conjecture, Lean or compression claim is made.